

UNIVERSITY of WASHINGTON

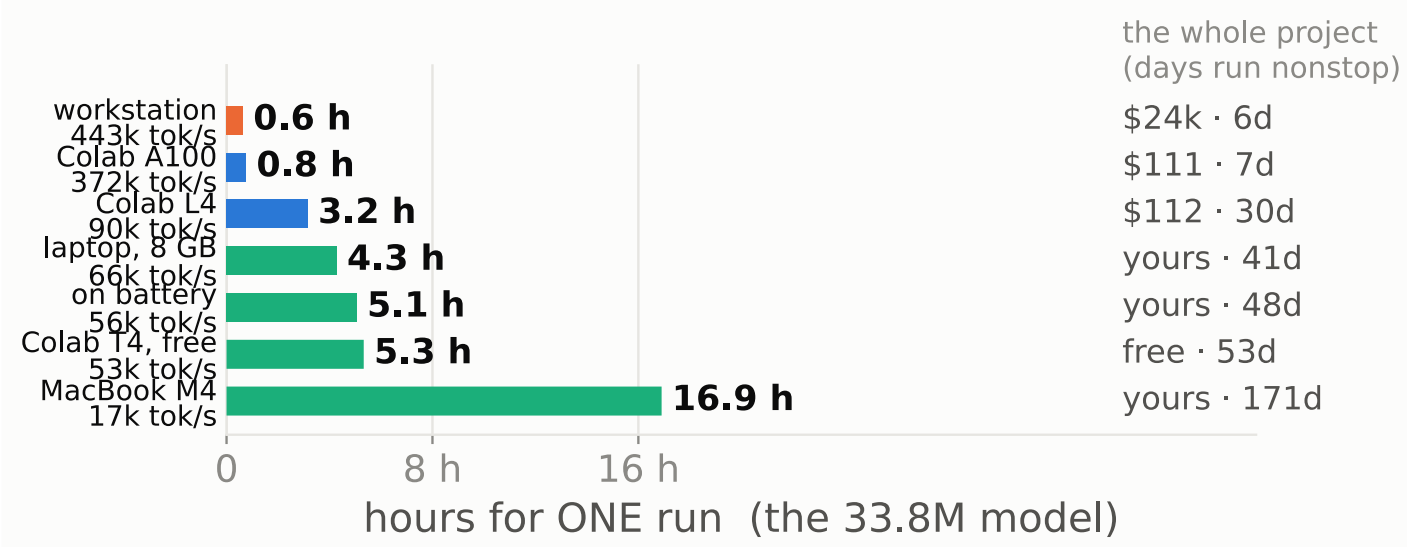
CSED 505: BUILDING A MODEL  
FACTORY



501 the statistics · 502 the mechanics · 503 the language stack · 504 scale — All four end when a model finishes training. None covers needing a hundred models and having to believe the differences between them.

Jeffrey Stall · Patrick Kwok · Leon Wan — CSED 504. This board is the machinery; the upper board is the experiment it served.

### You do not need the workstation



An 8 GB laptop GPU beats the free tier, even unplugged. The two paid Colab tiers bill the same — \$111 for 7 days against \$112 for 30 — so the faster tier buys time, not money. Electricity for all 148 GPU-hours was \$7; the workstation that ran them, \$24,000.

### Budget in tokens, not epochs

Epochs were right in 502–504: the dataset was fixed. Here it is the experiment — "40 epochs" gives one arm 256×.

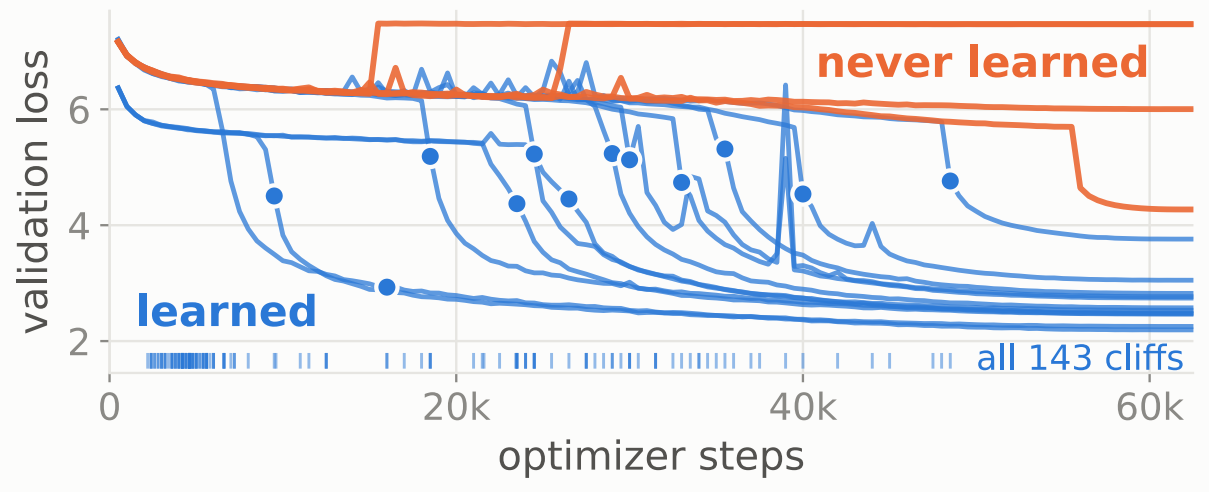
So fix tokens, and let epochs fall out. Every run is 1.024B tokens: the 4M corpus seen 256×, the 1,024M once.

Tokens ÷ tok/s = the wall clock. 1.024B ÷ 443k ≈ 38 min — the chart's top bar.

A truncated run is a different experiment. The schedule anneals to zero at the planned end — halfway is mid-schedule.

Our own records carried the bug. The epoch field reads 125; the truth: 16 passes.

### You cannot judge a run early



Yoruba, English and ten more, all on one axis. Every run that learns sits flat and then falls off a cliff — and no two fall at the same step: earliest 2,200, latest 48,500. So "still flat at step k" is evidence of nothing. 35 of 197 never learned.

### A hundred runs is a filing problem first

A study is declared as a grid — languages × rates × seeds — never typed run by run. The queue orders it longest-budget-first. Every run writes a record naming everything that changes the answer — settings, data fingerprint — after two runs differing only in vocabulary silently overwrote each other. "What have we tried?" is a query: 1,089 runs later, no collisions, no archaeology.

### A factory that makes one thing is not a factory

English is the ruler. Why: the one language where data was never the constraint. Learned: more data stops helping past 64M tokens — which Yoruba, having no more, could never show.

French, Indonesian, Mandarin are the control. Why: XLM-R knows these three well. Learned: they cost 1.04, 1.01, 0.95 — the penalty tracks coverage, not "African".

Twelve African languages are the gradient. Why: an anecdote needs a slope behind it. Learned: the split falls along coverage — Wolof, uncovered yet cheap at 1.31, the exception anecdotes hide.

Five languages got the whole sweep. Why: "one call per language" is only true if settings transfer. Learned: they do not — the rate four languages want destroys Igbo.

None of us speaks any of them. So the spread is the evidence — nine functions, one call per language, twelve in 48 minutes.

### When memory is a wall, and when it is only a speed limit

What has to fit is the model — 12–16 bytes a parameter. Our 98M: 1.6 GB. A 7B: 84–112 GB — no consumer card.

Everything above the floor is batch, and batch is optional. Accumulation runs a big step as smaller passes — same update, within 2%.

So an 8 GB laptop trains the step a 96 GB card trains. A 10 GB run folds into 6.1 GB at 28,027 tok/s.

Nobody on the chart beside this needed 80 GB. 96, 80, 24, 16, 8 GB — all ran the same step. The A100 sells speed, not room.

And Windows does not warn you. An oversized batch spills to system RAM: 28,027 → 5,075 tok/s, no error.

### What we made faster

2.07× on the four-cell benchmark — only 1.32× of it efficiency; the rest is a second card nobody had used. A third of the twenty were rejected, and a list of only the wins would be marketing rather than a log. Every row had to make the same experiment faster.

| WHAT CHANGED                          | MEASURED                       | KEPT?   |
|---------------------------------------|--------------------------------|---------|
| THE DATA PATH                         |                                |         |
| 1 · dataset on the card, no loader    | 13.8k → 18.5k img/s            | yes     |
| 2 · images as uint8, cast per batch   | 4× less memory, free cast      | yes     |
| 3 · augment on the GPU                | no PIL, no host copy           | yes     |
| 4 · tokenize once, up front           | 53 s vs. 85 min — 96×          | yes     |
| 5 · token dtype from the vocabulary   | 16k → 2 bytes a token          | yes     |
| PRECISION AND KERNELS                 |                                |         |
| 6 · bf16 autocast, not fp16           | 1.34× on sm_89, no loss-scale  | yes     |
| 7 · channels_last for CNNs, bf16 only | fp16 + it is 3.5× slower       | guarded |
| 8 · channels_last for ViTs            | a no-op on a transformer       | no      |
| 9 · TF32 for matmul and cuDNN         | free on Ampere and later       | yes     |
| 10 · cudnn.benchmark = True           | worth it when shapes are fixed | yes     |

### The factory, in numbers

- 197 pretraining runs, every one from scratch
- 892 fine-tuning runs on top of them, across two tasks
- 17 languages measured, 12 of them pretrained from scratch
- 148 GPU-hours, two cards, 71 kWh at the wall
- 62,500 steps a run — 1.024B tokens of updates, the unit that makes two machines comparable
- 0 identity collisions, because a vocabulary carries a hash

### The setting four languages want destroys the fifth

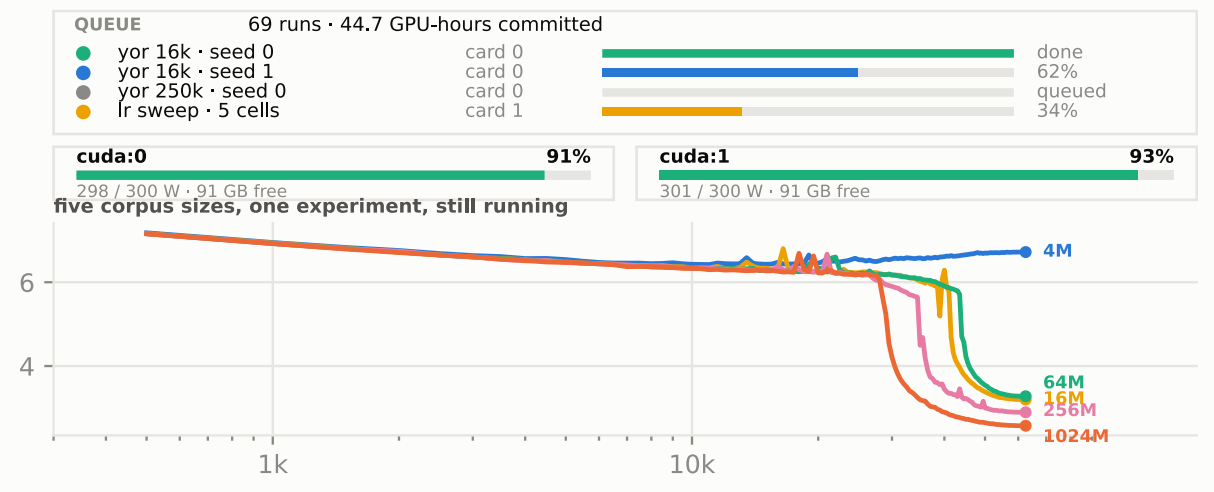
|         |     |       |       |                                         |       |       |
|---------|-----|-------|-------|-----------------------------------------|-------|-------|
| Yoruba  | ✗   | +0.22 | ★     | +0.02                                   | +0.19 | 🌙     |
| Swahili | ✗   | +0.55 | +0.09 | ★                                       | +0.06 | +0.16 |
| Nyanja  | ✗   | +0.70 | +0.12 | ★                                       | +0.05 | +0.64 |
| Hausa   | ✗   | +0.38 | +0.07 | ★                                       | +0.02 | +0.05 |
| Igbo    | 🌙   | +0.07 | ★     | ✗                                       | ✗     | ✗     |
|         | 1.5 | 3     | 5     | 7                                       | 8.5   | 10    |
|         |     |       |       | peak learning rate (×10 <sup>-4</sup> ) |       |       |

Five languages, six learning rates, sixty runs, each cell scored against that language's own best: ★ its winner, ✗ far behind it, 🌙 one seed of two. Four land at their best in the outlined column. Igbo collapses there — the same setting, a wasted night.

### A result has to beat your own noise

Run the same recipe twice and it lands in two different places. That is the seed, not the change you made. So we measured our own noise first and made every claim clear it by 2.27× — at three runs a side, no test can call anything significant. The spread is a property of each cell — re-measured, never a constant we reused. Two of our own claims did not clear it. We retired them.

### One screen, both cards, a rush order at 9 p.m.



The chart is five English corpus sizes separating while they run. On 9 August 44.7 GPU-hours sat across 69 runs and Patrick needed a sweep: a fleet pins to one card, so it took card 1 while the queue kept card 0. Twenty minutes to running, on a measured promise.

### What we do not claim, and what we would do next

Nobody on this team reads Yoruba — every judgement here is a benchmark number. The corpus is scraped web text nobody consented to. The audit gate prints 9 claims: 6 supported, 2 not, 1 underpowered. Next: a wall meter, a corpus audit, the evaluation we could not do.

### Sources

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  - Devlin et al. — BERT, NAACL 2019
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  - Dodge et al. — "Show your work," EMNLP 2019